BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 044 · Data Structures · Classes and Objects in C#

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a method parameter?

Q2. What is a breakpoint?

Q3. How does a queue differ from stack?

Q4. State the output of 1 AND 0.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — class

Recall the term — object

Recall the term — field

Recall the term — method

Recall the term — instance

New learning

Class ≠ instance — A class defines the fields and behaviour available to each object. new Student() constructs an instance; a separate new creates another independent object.

Worked example

One class, two independent objects

1. Identify class, objects and members — Badge is the class (blueprint). amber and blue are references to two objects . Each has its own Points field and can run the Add method.

2. Trace the state change — new Badge() creates each object with Points = 0 . amber.Add(2) changes amber’s field from 0 to 2; blue remains at 0 because it is a different instance.

3. Predict, then check the output — The output is 2,0 . A class definition does not create an object. A method call on one object does not automatically change the fields of another.

We Do · Practise with guidance

Different class, different data

```
using System;
Counter first = new Counter();
Counter second = new Counter();
first.Increase();
first.Increase();
Console.WriteLine(first.Value + "," + second.Value);

class Counter {
 public int Value = 0;
 public void Increase() { Value++; }
}
```

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

What are first.Value and second.Value ?

Trace both instances

Start with separate fields: first = 0, second = 0. Track which object receives each method call before revealing the result.

Independent state

first.Value = 2 and second.Value = 0 . Both calls act on first ; they do not change second .

Change the receiver

Continue from Example 1: second.Increase(); What values are stored now? Explain what second identifies.

Predict before checking

Which instance runs Increase() ? Which instance field changes?

Targeted update

first.Value = 2 ; second.Value = 1 . The variable second refers to the second created Counter instance.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply independently
Write C# to define a class Book with public string Title. [3]

Question 2

Apply independently
Instantiate a Book object named novel. [2]

Question 3

Apply independently
Explain why two objects of the same class may hold different field values. [2]

Exit Ticket · Answer before checking

True or false: defining a class automatically creates an object.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Named input to a method

Q2. A debugger stopping point

Q3. FIFO versus LIFO

Q4. 0

You Do pages · Independent questions

Q1. class Book { public string Title; }

Q2. Book novel = new Book();

Q3. Each object has its own instance state.

Exit Ticket · Model answer

False. A class describes the structure and behaviour of objects, but defining the class does not instantiate anything. An object exists only when code creates an instance, for example using new.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: